Federated Bee-Inspired Drone Hives for Fault-Tolerant, Non-Invasive Reconnaissance

Architecture, Simulation Baselines, and Preliminary Software Verification

Research manuscript / technical preprint draft, version 0.1 (14 October 2026). Not peer reviewed. Provisional authorship: Project researcher(s); confirm the byline and affiliation before submission. **Code and evidence:** https://github.com/Tybent18/distributed-drone-hive **Keywords:** multi-agent systems; swarm robotics; federated autonomy; network resilience; mobile infrastructure; archaeological reconnaissance.

Abstract

Aerial reconnaissance systems operating in difficult or access-constrained environments depend not only on capable individual vehicles but also on communications, safe refuge, replenishment, provenance, and recovery infrastructure. We introduce *Drone Hive*, a conceptual federation of independently governed, bee-inspired drone colonies using distinct solar/battery energy roots (N0), Queen-led primary hives (N1), downstream staffed modular nests (N2+), and temporary finite-energy micro-nests called eggs. We distinguish the construction-parent graph, communications mesh, and energy-accounting relationships. A conditional bypass hypothesis - N1 to E1 to E2 to N3 after permanent N2 failure - is formalized as a falsifiable design target, not a proven guarantee. We separate two evidence levels. The frozen September 2026 Stage One simulation compared carrier-only, fixed-relay, and adaptive-frontier infrastructure in five-seed trials (24 simulated drones, 240 steps). Fixed relays achieved 51.58% mean mapped area and 74.17% final drone survival; the adaptive policy achieved 100% nominal final drone survival but 0% with the specified first-relay failure. Stage Two implements role/authority invariants, finite docking reservations, abstract energy conservation, obstacle-aware radio geometry, and traceable synthetic records. A verified October 2026 workflow passed 39 software tests on Python 3.10 and 3.12 and completed two 210-row exploratory diagnostics. These data do **not** establish physical feasibility, calibrated archaeological measurement, or comparative superiority of the proposed egg deployment. We conclude with a preregistered path toward matched-resource evaluation and hardware-informed validation.

1. Introduction

Remote reconnaissance and documentation of hazardous, collapsed, or inaccessible terrain expose a systems-level problem. A swarm may navigate locally yet still fail to deliver useful data when battery stores are depleted, a relay is lost, a docking station is overfull, or a repair unit cannot reach the fault. Swarm robotics emphasizes distributed coordination and resilience, while aerial-swarm research further highlights limited payloads, flight endurance, communications, and mission safety [1,2].

Drone Hive proposes persistent and temporary support infrastructure for mission-specific aircraft. Its bee-inspired naming refers to **organizational roles**, not an implemented biological algorithm: Queens coordinate a colony, Handlers maintain infrastructure, Workers conduct survey tasks, nests provide physical services, and eggs are constrained emergency modules. The intended application is **non-contact reconnaissance**. Archaeological views in the project are inspired by Göbekli Tepe, not an authorized or measured reconstruction of that World Heritage site [4].

This paper contributes a research architecture, the preserved historical simulation comparison, the status of preliminary engineering verification, and a protocol to test whether reserve micro-nests improve safety or mission continuity **after accounting for their cost**. It is deliberately neither a certified UAV safety case nor a positive treatment-effect study.

2. System Architecture and Falsifiable Hypotheses

2.1 Independent command

Each colony has a Node 0 energy root, exactly one Queen at Node 1, and exactly three multifunction Node 1 Handlers. Each commissioned downstream nest N2+ has one dedicated Relay Handler, one Retriever Handler, and mission-dependent specialist Workers. No Queen is placed downstream. Authorized assistance between colonies can carry communication and rescue requests, but must not transfer control. Temporary local delegation to an available Node 1 Handler uses an authority epoch in the software prototype, without claiming proven distributed consensus during a network partition.

2.2 Separate graphs and resources

We conceptually distinguish:

- **G_build:** physical commissioning and parent/child transport dependencies;
- **G_comm(t):** powered nodes and viable radio links, including optional egg relays;
- **G_energy(t):** sources, finite storage, charging, transfers, and dissipative losses;
- **G_logistics(t):** viable movement paths and transport capacities (not yet fully implemented).

A radio edge is **not** evidence of physical electrical energy transfer. The Stage Two communication model uses configurable geometric range and simple opaque rectangular line-segment exclusion. It omits measured frequency-dependent path loss, interference, packet drops, bandwidth congestion, and multipath. The energy model tracks abstract quantities and hypothetical parent transfer efficiency rather than physical power engineering.

2.3 Specialized bees, hives and reserve eggs

Workers specialize in scouting, depth/LiDAR mapping, close-range noncontact structural imaging, or environmental sensing. The prototype identifies sensor classes and emits synthetic observation metadata; it does not generate measured video, photogrammetric products, LiDAR point clouds, or field-calibrated science data. Modular nest construction and autonomous Retriever transport remain research objectives.

Eggs are positioned at commissioned nest midpoints and offer a subset of a nest's services: temporary radio relay, finite charging/refuge, and buffer space. Consider the nominal corridor **N1 - E1 - N2 - E2 - N3**. Following N2 failure the hypothesized **N1 to E1 to E2 to N3** bypass is permitted only when E1/E2 are active and powered, all three radio segments are reachable, and safety resources remain. A depleted pod, blocked route, damaged neighbor, or insufficient radio range invalidates the bypass.

2.4 Safe failure behavior

Energy, structural, destruction, or blocked-access failures lead to affected-colony conservative recall to a reservable dock or a safe shelter response. Comms-only anomalies trigger diagnosis and a bounded timeout rather than immediate physical-destruction assumptions. A neighboring colony may assist but retains its own sovereign command hierarchy. Physical servicing needs authorized Retriever action and finite equipment resources; those costs are not yet realistically simulated.

3. Methods and Evidence Boundaries

3.1 Stage One archived simulation

The **frozen 15 September 2026** Stage One baseline tested carrier-only, fixed-relay, and adaptive-frontier policies. Five random seeds, 24 drones, and 240 discrete simulated steps were used per available condition. A relay-failure condition was available only for policies that deploy relays; **carrier-only is not assigned an invented failure outcome**. Simulated radial energy fields, activity/survival, exploration and connectivity are dimensionless and not physical radio or battery measurements.

The audit bundle includes source manifest, run summaries, step telemetry, aggregate CSVs and plots [5]. It is preserved rather than overwritten to make the policy results reproducible.

3.2 Stage Two software verification

Stage Two introduces independent colony geometry, graph-based radio, midpoint pods, docking slot reservations, synthetic Worker record identifiers/checksums, unavailable-Queen checks and explicit abstract energy-event bookkeeping. Python 3.10/3.12 CI run **#38653201008** reports **39 passing tests** for the modeled software assertions [7].

Two Stage Two engineering exploratory matrices each generated **210 rows**: (a) ten deterministic seeds × seven fault scenarios × three phases, and (b) ten paired seeds × seven architectural C0–C6 configurations × three scenarios. The latter arms have **unmatched effective staffing and hardware budgets** (e.g., deactivated nests may retain scripted Workers), so their outputs are **mechanism/debugging diagnostics**, not unbiased estimates of treatment efficacy. Workflow-generated CSV artifacts can expire and are not a permanent frozen Stage Two release.

3.3 Planned controlled intervention study

A planned preregistered design [6] compares C0 minimal colony core, C1 linear permanent nests, C2 branched network without eggs, C3 root-stored eggs, C4 midpoint non-relaying logistics/refuge eggs, C5 midpoint radio/refuge eggs, and C6 C5 with cross-colony assistance. Conditions must equate starting hardware, energy, valid drone staffing, terrain and mission tasks, with full accounting for equipment costs. Collect an initial ~10 development seeds and **at least 30 separate paired evaluation seeds**, adjusted by pilot statistical precision analysis. Prespecified primary endpoints will be **true safe-refuge success** and **ground-truth-verified mission coverage under failure**. Report paired effect sizes, intervals, unsuccessful runs and sensitivity to failures, radio conditions and battery availability.

4. Archived Quantitative Results

Table 1. Frozen Stage One aggregate outcomes (means across five seeds; available simulator conditions only).

Infrastructure policy	Test condition	Mapped area	Final simulated drone survival	Connected hive fraction
Carrier only	Nominal	43.84%	0.00%	100.00%
Fixed relay	Nominal	51.58%	74.17%	100.00%
Fixed relay	First-relay failure	51.58%	74.17%	83.33%
Adaptive frontier	Nominal	36.16%	100.00%	100.00%
Adaptive frontier	First-relay failure	36.14%	0.00%	20.00%

The historical comparison supports the conclusion that *in this model* relay placement and topology can greatly affect resilience. Fixed relays maintained their observed final survival under the one tested first-relay fault; the adaptive deployment failed to retain drones under that fault. The data are insufficient to conclude that fixed placement is always best, or that the Stage Two egg concept has been validated. The outcome variability and five-seed sample size restrict inference.

Stage Two verified output. The most recent validated workflow passed 39 software tests on each supported Python version and generated 210 fault-diagnostic rows plus 210 C0–C6 engineering-ablation rows. These are **observation counts**, not 420 independent experimental trials or evidence of improved mission effectiveness.

5. Discussion and Limitations

The signature question is not simply whether adding another graph vertex allows a route. It is whether a physically positioned, powered, finite-capacity micro-nest can be deployed and serviced within mission constraints, and whether any benefit survives comparison against equally provisioned redundant permanent networks and root-stored alternatives. This remains **unproven**.

Source-of-truth distinctions are essential. **Scripted React/Three.js GIFs illustrate scenarios; they are not Python-model telemetry.** Sensor metadata hashes are computed over synthetic event strings and do not validate physical observations. Conserved *dimensionless* energy does not establish realistic charging efficiency, battery endurance, or wireless power. Geometric obstacles provide a useful engineering rejection test but not a calibrated communications link budget. Timer-based repair does not estimate Retriever transit time, loaded payload mass, repairs, material inventory or docking clearance.

The archived five-seed Stage One measurements do not include the new Queen/Handler roles or emergency eggs and must never be pooled with Stage Two. A future held-out, resource-normalized comparative study must report not only successful cases but unrecoverable faults and data loss. The simulation cannot support claims about operation at a protected archaeological site without permits, aviation and spectrum compliance, and safety review.

6. Research Agenda

1. **Construction and logistics:** develop staged deployment with finite chassis inventory, carrying mass, route geometry, validation gates, and Retriever missions. 2. **Energy and docking:** distinguish locally generated supply, finite transfers, queueing capacity, charge durations and cell deterioration using sourced physical parameters. 3. **Communications:** extend LOS geometry to obstruction, bandwidth, packet loss, congestion and network partitions; instrument confirmed route recovery delays. 4. **Data quality:** compare delivered, unique and calibrated synthetic observations to independently generated ground-truth scenes, rather than equating bytes with mapping accuracy. 5. **Controlled evaluation:** run matched C0–C6 arms on held-out terrain/fault seeds; report paired differences and uncertainty intervals. 6. **Visualization traceability:** replay serialized Python event streams in the WebGL frontend, preserving source run IDs. 7. **Hardware feasibility:** pursue component benchmarking, safety review and authorized field validation only after modeled assumptions are tested.

7. Conclusion

Drone Hive presents a falsifiable federated drone-colony design with independent command, specialized mission vehicles, modular permanent nests and finite midpoint contingency micro-nests. Archived Stage One software results reveal strong sensitivity to infrastructure topology and a brittle relay-failure mode in an adaptive deployment. Stage Two has improved software safety constraints and diagnostic execution, but **the proposed egg-bridge performance advantage has not been scientifically established**. The next meaningful contribution is a balanced, reproducible comparative trial using physically motivated costs and verifiable mission outputs.

Data and Software Availability

The project code, historical Stage One evidence and current state of development are available at https://github.com/Tybent18/distributed-drone-hive [5]. The Stage Two protocol, code changes and preliminary CI evidence were developed in PR #4 [6,7]. CI diagnostic artifacts may expire. No human-subject data, physical flights or protected-site surveys were collected.

Declarations

Authorship, affiliation, funding, contributions, competing interests and journal: to be confirmed before submission. **Peer-review status:** Technical preprint draft, not submitted or reviewed. **Ethical / operational limits:** Software-only research; no claimed aviation or archaeological-site authorization.

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